

WASHINGTON WARREN FIELD, NC, USA

WMO: 746925

Lat: 35.570N

Lon: 77.050W

Elev: 38

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 03741

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.8	25.2	-10.1	3.2	43.9	-0.5	5.4	39.3	20.7	58.4	18.4	53.7	3.8	350	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	94.5	78.3	92.6	77.0	90.2	76.4	81.8	88.6	80.3	87.3	78.8	86.0	7.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.6	159.1	86.2	78.7	149.4	84.3	76.8	139.6	83.2	45.5	89.2	43.6	86.2	42.4	86.4	91.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	15.7	12.9	15.4	98.1	3.3	2.2	13.0	99.7	11.1	101.0	9.3	102.2	6.9	103.9		
(5)			12.0	84.0	4.9	2.9	8.5	86.1	5.6	87.8	2.9	89.4	-0.7	91.5	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.6	43.3	46.0	52.6	62.2	70.1	77.6	80.7	78.9	73.5	63.8	53.6	47.6
	DBStd	15.29	11.09	9.58	9.80	8.32	7.06	5.18	4.31	4.54	5.27	8.10	8.81	9.63
	HDD50	765	268	174	89	7	0	0	0	0	0	5	59	163
	HDD65	2819	674	536	400	150	34	1	0	0	5	120	354	545
	CDD50	5360	61	62	169	373	623	828	952	897	705	434	168	88
	CDD65	1942	3	5	15	66	192	379	487	432	261	84	13	5
	CDH74	17597	6	27	130	588	1632	3557	5002	4048	1925	593	71	18
	CDH80	6256	0	2	13	124	524	1368	2048	1522	534	117	4	0
Wind	WSAvg	5.2	5.3	6.0	6.4	6.6	5.8	5.0	4.9	4.1	4.9	4.4	4.6	4.7
Precipitation	PrecAvg	48.9	3.7	3.0	4.3	3.1	4.0	4.3	5.2	6.3	3.7	3.3	3.2	
	PrecMax	65.6	6.7	6.6	6.9	5.7	8.1	11.0	9.5	11.5	26.2	12.2	8.6	8.0
	PrecMin	36.0	1.3	1.1	1.5	1.3	1.7	0.3	2.4	2.6	0.9	0.0	0.7	0.9
	PrecStd	7.9	1.5	1.5	1.4	1.3	1.7	2.4	1.6	2.2	5.7	2.5	2.0	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.5	78.7	81.6	87.1	93.5	97.2	98.9	95.4	91.4	87.8	80.7	77.1
		MCWB	66.3	66.9	63.3	67.3	75.1	75.5	77.6	80.3	76.3	75.0	67.2	69.3
	2%	DB	71.2	72.3	78.0	83.0	89.4	93.1	94.7	93.0	88.0	82.7	75.4	72.2
		MCWB	64.1	61.6	63.3	66.2	73.6	76.1	79.5	78.2	74.4	70.7	64.6	65.1
	5%	DB	64.4	67.1	73.3	81.0	85.6	90.9	92.6	90.5	85.2	80.7	72.4	68.0
		MCWB	58.2	59.6	61.1	65.8	71.8	76.2	78.0	77.1	73.0	69.4	64.2	63.3
	10%	DB	61.4	63.3	69.7	77.7	82.1	87.8	89.9	87.8	82.5	77.3	68.5	63.8
		MCWB	55.3	54.6	60.3	65.3	70.8	75.8	77.1	76.4	72.4	67.7	62.2	58.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.9	67.7	70.3	76.4	81.8	84.0	84.7	83.6	80.0	77.3	72.0	70.6
		MCDB	73.6	75.4	76.4	80.4	88.3	89.2	94.7	94.2	84.2	83.1	77.3	74.9
	2%	WB	64.4	64.8	66.9	72.8	77.7	81.2	81.8	80.7	77.8	74.8	68.4	66.8
		MCDB	69.3	70.6	74.3	77.8	83.1	85.3	89.4	89.5	83.8	80.8	72.6	70.4
	5%	WB	60.3	61.3	64.5	70.0	75.0	79.2	80.2	79.0	76.3	72.7	66.0	63.4
		MCDB	64.5	66.5	70.3	76.4	80.8	84.2	88.5	87.3	82.7	78.2	71.0	66.7
	10%	WB	56.6	56.4	61.8	67.6	73.1	77.7	78.7	77.8	74.9	69.9	62.5	59.4
		MCDB	60.4	61.3	67.7	75.0	79.7	84.6	86.8	85.4	81.1	75.1	67.4	63.6
Mean Daily Temperature Range	5% DB	MDBR	19.0	20.2	21.1	21.5	19.3	17.7	16.6	16.4	16.7	19.6	21.3	19.3
		MCDBR	23.7	25.3	25.4	24.2	22.2	20.1	18.5	18.1	19.2	20.7	23.1	21.2
		MCWBR	18.3	17.4	14.9	12.2	10.0	9.0	8.7	8.9	9.2	12.1	15.4	16.5
	5% WB	MCDBR	21.7	23.7	21.0	20.4	19.0	17.2	16.5	15.9	14.9	16.1	19.2	19.4
		MCWBR	19.0	19.0	15.4	13.8	10.7	9.7	8.8	8.9	8.7	11.1	15.3	16.6
Clear-Sky Solar Irradiance	taub	0.310	0.326	0.353	0.395	0.443	0.497	0.526	0.515	0.416	0.362	0.329	0.316	
	taud	2.531	2.480	2.432	2.320	2.222	2.115	2.048	2.070	2.348	2.480	2.526	2.546	
	Ebn at Noon	284	291	291	282	269	253	245	245	266	274	274	274	
	Edh at Noon	25	30	34	40	45	50	53	51	37	30	25	24	
All-Sky Solar Radiation	RadAvg	814	1041	1366	1734	1911	2011	1915	1735	1435	1180	900	722	
	RadStd	52	111	93	117	140	108	105	99	100	141	66	57	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (61 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+281	+198

Nomenclature: See separate page